

Imperial School of Banking & Management Studies

Session Plan – PGDM 2025–27 | Semester III

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Course Title	Agentic AI & Advanced Analytics in Finance
Course Code	PGDM-SEM3-SPEC-AIFINANCE
Specialization	Finance (Elective / Specialization Subject)
Programme	PGDM 2025–27 Batch — Semester III
Faculty	TEJAS JADHAV, CFA, FRM MMS (SIMSREE) Mumbai
Total Teaching Hours	30 Hours (10 Sessions × 3 Hours each)
Session Plan Submitted On	June 2026
Academic Year	2025–26 (Semester III)

About the Course

Agentic AI & Advanced Analytics in Finance is a 30-hour specialization course for Semester III PGDM students at ISBMS, designed to bridge theoretical finance and contemporary AI-driven industry practice. The course equips students with production-grade skills in demand at Global Capability Centres (GCCs) and financial institutions across India and globally. Students explore agentic AI architectures, prompt engineering, Retrieval-Augmented Generation (RAG) pipelines, algorithmic trading, credit and market risk modelling (PD models, Monte Carlo VaR), regulatory compliance automation (Basel III/IV), and fund administration workflows. Drawing from Harvard Case Method, NUS sandbox-first learning, GARP's FRM framework, and CFA Institute standards, each module culminates in a deployable project artifact that builds a live portfolio. The capstone simulates a real GCC hiring panel, ensuring students are placement-ready upon completion.

Word count: ~130 words

Course Objectives (Bloom's Taxonomy)

- To **recall** key concepts of Large Language Models, AI agent architectures, and financial data APIs as applied in financial services operations. **L1 – Remember**

2. To **explain** the principles of Retrieval-Augmented Generation (RAG), Model Context Protocol, and multi-agent orchestration within the context of financial workflow automation. L2 – Understand
3. To **apply** prompt engineering, Python programming, and quantitative modelling techniques in hands-on lab sessions to solve real-world financial problems. L3 – Apply
4. To **analyse** risk metrics (VaR, PD), regulatory compliance requirements (Basel III/IV), and algorithmic trading performance using AI-assisted analytical tools. L4 – Analyse
5. To **evaluate** enterprise AI governance frameworks, model risk management policies, and the regulatory and ethical implications of deploying AI in financial institutions. L5 – Evaluate
6. To **design** an integrated financial AI system—encompassing a fund administration module, risk analytics engine, and compliance report generator—demonstrating placement-ready skills for GCC and financial sector roles. L6 – Create

Session-Wise Teaching Plan (10 Sessions × 3 Hours = 30 Hours)

Sess.	Hour	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
Session 1 Module 1: Foundations of Agentic AI in Financial Services 3 Hours					
1	Hour 1	LLM Fundamentals & Financial AI Overview: Transformer architecture, attention mechanism, tokenization; evolution of AI in finance from rule-based → ML → deep learning → agentic systems; Bloomberg GPT, FinBERT, Claude API applications in financial services	Recall LLM concepts; explain financial NLP applications (L1, L2)	Lecture; Concept Quiz; Group Discussion: "How has AI changed investment banking?"	Jadhav, <i>Claude AI for Finance Professionals</i> (Ch.1); GARP FRM Study Notes – Technology Risk
	Hour 2	Agentic AI Architecture: Observe–Plan–Act–Reflect agent loop; tool definitions and function calling;	Explain agentic architecture; apply agent design to financial use-cases (L2, L3)	Lecture + Live Demo; Case: "Morgan Stanley GPT-4 Wealth	LangChain Docs; Anthropic API Reference; Hilpisch (2020) Ch. 2

Sess.	Hour	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		agent memory types (short-term, episodic, semantic); financial data APIs — NSE, BSE, Yahoo Finance, RBI DBIE		Advisor Program"	
	Hour 3 – Lab	In-Class Lab: Build a structured financial data extraction prompt using the Claude API and Python (VS Code) — extract income statement line items from unstructured text; test with 3 different annual report excerpts	Apply prompt engineering to extract structured financial data (L3)	In-Class Python Lab (VS Code + Claude API); Instructor Demo; Peer Discussion	Anthropic Claude API Documentation; LangChain Python Docs
Session 2 Module 2: Prompt Engineering & RAG for Financial Analysis 3 Hours					
2	Hour 1	Prompt Engineering for Finance: Zero-shot, few-shot, chain-of-thought (CoT) prompting; role prompting ("You are a CFO..."); output formatting (JSON, markdown tables); hallucination risks and mitigation strategies	Apply prompt patterns; evaluate quality of AI financial outputs (L3, L5)	Lecture + Live Prompting Exercise; Pair Activity: compare prompt outputs for identical query	Jadhav, <i>Claude AI for Finance Professionals</i> (Ch.3–4); White et al. (2023) ArXiv: 2302.11382
	Hour 2	RAG Pipeline Construction: Vector databases (ChromaDB, Pinecone); document chunking	Explain RAG pipeline components; design retrieval strategy for financial	Lecture; Architecture Diagram Workshop; Group Activity: redesign five	LlamaIndex Documentation; Lewis et al. (2020) NeurIPS ArXiv:2005.11401

Sess.	Hour	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		strategies for financial documents; embedding models (Sentence Transformers); retrieval and re-ranking; advanced prompting: self-consistency, tree-of-thought	documents (L2, L3)	failing financial prompts	
	Hour 3 – Lab	In-Class Lab: Build a Q&A bot on a 200-page NSE IPO Prospectus using LlamaIndex + ChromaDB + Claude API (VS Code); test with 10 investor questions; evaluate retrieval accuracy	Design and deploy a RAG pipeline for unstructured financial documents (L6)	In-Class Python Lab (VS Code + LlamaIndex + ChromaDB); Industry Case: "SEBI DRHP Analysis using RAG"	LlamaIndex Python Docs; ChromaDB Documentation
Session 3 Module 3: Model Context Protocol & Agent Orchestration 3 Hours					
3	Hour 1	Model Context Protocol (MCP): Protocol specification; tool definitions and JSON schemas; context window management for long financial documents; memory management (summarization, rolling context); financial context design patterns — credit analyst agent, portfolio manager agent	Explain MCP architecture; apply context design in financial scenarios (L2, L3)	Lecture; Architecture Diagram Workshop; Simulation: design a credit analyst MCP agent schema	Anthropic MCP Specification (2024); Jadhav, <i>Claude AI for Finance Professionals</i> (Ch.6)
	Hour 2				

Sess.	Hour	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		Multi-Agent Frameworks: CrewAI and LangGraph introduction; agent roles (researcher, analyst, writer, reviewer); sequential vs. hierarchical vs. parallel workflows; inter-agent communication; designing agent crews for financial due diligence and investment research	Design multi-agent pipelines; evaluate architectural choices for financial workflows (L6, L5)	Lecture; Live Demo (CrewAI multi-agent crew); Architectural Design Exercise in pairs	CrewAI Documentation; Wu et al. (2023) AutoGen ArXiv:2308.08155
	Hour 3 – Lab	In-Class Lab: Build a 3-agent Investment Memo pipeline — Research Agent (data gathering) → Analysis Agent (financial interpretation) → Writing Agent (memo drafting); run on a real NSE-listed company; peer review memo quality	Create and evaluate a multi-agent financial research pipeline (L6, L5)	In-Class Python Lab (VS Code + CrewAI + Claude API); Peer Review Panel	CrewAI Python Docs; Anthropic API Reference
Session 4 Module 4: Algorithmic Trading & Execution Strategies 3 Hours					
4	Hour 1	Market Microstructure & Trading Signals: Order book dynamics, bid-ask spread, order types (market, limit, stop-loss, IOC); trading signal categories —	Explain algorithmic trading concepts; analyse signal performance metrics (L2, L4)	Lecture; Simulation Exercise (virtual order book); Case Excerpt: "Renaissance Technologies Medallion Fund"	Chan (2013) Chs. 1–3; SEBI Circular on Algorithmic Trading (2012)

Sess.	Hour	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		momentum (RSI, MACD), mean-reversion (Bollinger Bands), pairs trading (cointegration); SEBI algorithmic trading and co-location guidelines			
	Hour 2	Backtesting Methodology & Case Study: Walk-forward vs. in-sample/out-of-sample testing; overfitting and data snooping bias; performance metrics — Sharpe ratio, Sortino ratio, Maximum Drawdown; transaction cost modelling; Case Discussion: "Knight Capital Group \$440M algorithmic trading failure – August 2012"	Analyse backtesting results; evaluate strategy robustness and regulatory risk (L4, L5)	Case Discussion; Quantitative Exercise (compute Sharpe and Max Drawdown); Peer Analysis	Hilpisch (2020) Ch. 7; SEBI Order on Knight Capital India equivalent
	Hour 3 – Lab	In-Class Lab: Build an AI trading bot using Backtrader + Python (VS Code) implementing a momentum strategy on NSE-listed stocks with 2 years of data; generate a full backtested performance report (Sharpe, CAGR, Max Drawdown); NLP-based signal	Apply AI to trading strategy design; evaluate backtested Sharpe ratio and drawdown (L3, L4)	In-Class Python Lab (VS Code + Backtrader + yfinance); Peer Performance Review	Backtrader Documentation; yfinance Python Library; QuantLib Python Docs

Sess.	Hour	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		generation from financial news			
Session 5 Module 5: Risk Modelling – PD Models & Monte Carlo VaR 3 Hours					
5	Hour 1	Credit Risk & PD Models: PD, LGD, EAD components; Basel II IRB approach; ML-based PD model development using logistic regression, decision trees, and XGBoost; feature engineering from financial ratios; model validation — Gini coefficient, KS statistic, AUC-ROC; SHAP for explainability; Case: "Fintech lender PD model failure and regulatory action"	Apply ML models to credit risk; analyse model performance metrics (L3, L4)	Lecture; Python Demo (scikit-learn); Case Discussion; SHAP Visualization Demo	Basel Committee (2006) Basel II; Siddiqi (2017) <i>Intelligent Credit Scoring</i> Ch. 4
	Hour 2	Market Risk & Monte Carlo VaR: Historical simulation vs. parametric vs. Monte Carlo compared; 10,000-path Monte Carlo simulation for multi-asset equity portfolio; Conditional VaR (CVaR) / Expected Shortfall (ES); risk decomposition by asset; Basel III VaR thresholds (95%, 99%); Assignment	Design Monte Carlo VaR model; evaluate risk outputs against Basel thresholds (L6, L5)	Lecture; Industry Simulation (multi-asset portfolio VaR); Risk Interpretation Workshop	GARP FRM Part II – Market Risk; Hull (2022) Ch. 21–22

Sess.	Hour	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		introduced this session			
	Hour 3 – Lab	In-Class Lab: Build a Monte Carlo VaR engine in Python (VS Code + NumPy) for a 5-stock NSE portfolio; run 10,000 simulation paths; compute 95% and 99% VaR and CVaR; generate AI-written risk commentary using Claude API; output: formatted PDF report	Implement and interpret Monte Carlo VaR; produce AI-enhanced risk report (L6, L3)	In-Class Python Lab (VS Code + NumPy + Claude API + reportlab); Peer Report Review	NumPy Documentation; Anthropic Claude API; GARP FRM Study Materials
Session 6 Module 6: RegTech & Automated Basel III/IV Compliance 3 Hours					
6	Hour 1	Regulatory Framework – Basel III/IV: CET1, Tier 1, Total Capital adequacy ratios; Liquidity Coverage Ratio (LCR) and Net Stable Funding Ratio (NSFR); leverage ratio; Basel IV FRTB overview; RBI implementation circular for Indian banks; AI applications in continuous compliance monitoring; AxiomSL and OneSumX platforms	Recall Basel III/IV framework; explain AI's role in regulatory compliance (L1, L2)	Lecture; Regulatory Simulation (AxiomSL-style capital calculation template); Guest Video Segment	BIS Basel Committee (2017); Deloitte RegTech Universe Report (2023)
	Hour 2	AML, KYC & Regulatory	Analyse RegTech	Case Discussion;	EY RegTech Report: "AI in Regulatory

Sess.	Hour	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		Reporting Automation: AML transaction monitoring using ML; KYC document automation using NLP (Aadhaar/PAN OCR); SAR auto-generation; FATF Recommendations; RBI guidelines on AI in banking; Case: "HDFC Bank ₹1,000 crore RBI penalty – KYC compliance failure analysis"	automation use-cases; evaluate regulatory and operational risk (L4, L5)	AML Detection Simulation; NLP Demo (document classification for KYC)	Reporting" (2023); FATF Recommendations (2023)
	Hour 3 – Lab	In-Class Lab: Auto-generate a Basel III Pillar 3 disclosure document from structured bank data using Claude API and Python (VS Code + python-docx); extract capital, risk, and liquidity metrics; produce formatted Word document matching RBI Pillar 3 template; peer review output	Design automated compliance reporting pipeline; evaluate output accuracy (L6, L5)	In-Class Python Lab (VS Code + Claude API + python-docx); Peer Review of generated disclosure	Anthropic Claude API Documentation; RBI Pillar 3 Disclosure Template
Session 7 Module 7: Enterprise AI Governance, Security & Cost Management 3 Hours					
7	Hour 1	AI Governance & Model Risk Management: Federal Reserve SR 11-7 framework; model documentation,	Evaluate AI governance frameworks; analyse regulatory risk in AI	Case Discussion; Governance Blueprint Workshop; SHAP Visualization	Federal Reserve SR 11-7; FSB (2022) "AI and ML in Financial Services"

Sess.	Hour	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		independent validation, ongoing monitoring; SHAP and LIME for explainability; algorithmic bias and fairness in credit, insurance, and hiring; Case: "Goldman Sachs Apple Card gender bias – CFPB investigation 2019"	deployment (L5, L4)	Demo in Python	
	Hour 2	AI Security, Ethics & Cost: EU AI Act classification for financial services; India DPDP Act 2023; prompt injection attacks (direct and indirect); OWASP LLM Top 10; TCO modelling — API costs, compute, integration, maintenance; cost optimization strategies (caching, compression, batching); Case Study conducted this session	Evaluate security and cost trade-offs of enterprise AI deployment (L5, L4)	Red-Team Security Exercise (live prompt injection); TCO Spreadsheet Modelling; Regulatory Comparison Table	OWASP LLM Top 10 (2023); Jadhav, <i>Claude AI for Finance Professionals</i> (Ch.8)
	Hour 3 – Lab	In-Class Lab: Design an AI Governance Blueprint for a bank deploying an LLM-based credit decision system — covering model risk, data privacy, bias testing, and incident response;	Design an AI governance blueprint; evaluate compliance with SR 11-7 and EU AI Act (L6, L5)	In-Class Workshop (Excel TCO + Claude API + Word Blueprint); Peer Presentations	SR 11-7 Framework; EU AI Act (2024); DPDP Act India (2023)

Sess.	Hour	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		build TCO model in Excel; group presentations (5 min each)			
Session 8 Module 8: Fund Administration – NAV & Transfer Agency Automation 3 Hours					
8	Hour 1	Fund Administration Lifecycle: Fund types (mutual funds, hedge funds, PE/VC, AIFs); trade capture, portfolio valuation (mark-to-market, IFRS 9), corporate actions processing; NAV calculation steps; investor reporting; AI applications at Opus Fund Services, HC Global, SS&C GlobeOp, Apex Group; Case: "Opus Fund Services AI-powered fund automation GCC Pune"	Explain fund administration workflows; apply AI to NAV calculation process (L2, L3)	Lecture; Industry Case Discussion; Role-Mapping Exercise (identify AI insertion points in fund lifecycle)	IOSCO Principles for Fund Valuation (2013); Citi: "AI in Fund Administration" (2023)
	Hour 2	Transfer Agency & KYC Automation: Investor onboarding (AML/KYC), subscription/redemption processing, dividend distribution; Aadhaar-based e-KYC and PAN verification using NLP; SEBI mutual fund KYC	Evaluate KYC automation accuracy; analyse transfer agency workflow efficiency (L5, L4)	Lecture; Role-Play Exercise (investor onboarding simulation); KYC NLP Demo	KFINTECH Documentation; SEBI Circular on KYC Registration Agencies (2022)

Sess.	Hour	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		regulations 2023; KFINTECH and CAMS platforms; STP (Straight-Through Processing) and error reconciliation			
	Hour 3 – Lab	In-Class Lab: Build a Python-based NAV calculation engine (VS Code) for a 10-holding equity mutual fund; auto-generate a daily fund manager commentary and NAV report using Claude API; implement reconciliation check; output: formatted Excel/ PDF report matching SEBI disclosure format	Design an automated NAV reporting system; evaluate output quality (L6, L4)	In-Class Python Lab (VS Code + Claude API + openpyxl + reportlab); Peer Review of NAV report	Anthropic Claude API; openpyxl Documentation; SEBI NAV Disclosure Format
Session 9 Module 9: Market Commentary & Valuation Dashboards 3 Hours					
9	Hour 1	Financial NLP & Sentiment Analysis: FinBERT for financial sentiment; earnings call transcript mining and key phrase extraction; Bloomberg NLP use-cases; news-based event-driven signals; NLP pipeline — NSE announcement collection → preprocessing → FinBERT scoring	Apply NLP models to market analysis; analyse sentiment signals from financial text (L3, L4)	Lecture; Live Demo (FinBERT on NSE earnings transcript); Case Discussion	Yang et al. (2020) FinBERT IJCAI 2020; Bloomberg NLP Research

Sess.	Hour	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		→ AI-generated equity research commentary; Case: "Bloomberg Intelligence AI-generated earnings summaries"			
	Hour 2	Valuation Modelling with AI: DCF model automation using Claude API — revenue projection, WACC, terminal value; Comparable Company Analysis (CCA) — extract and compare EV/ EBITDA, P/E, P/BV multiples; ESG data integration in equity valuation; Damodaran valuation database application; live valuation exercise on NSE-listed company	Design AI-assisted valuation models; evaluate output accuracy against manual benchmarks (L6, L4)	Modelling Lab (Python-based DCF + Claude API); Live Company Valuation Exercise	Damodaran (2012) <i>Investment Valuation</i> Ch. 2; Damodaran Online Data; Anthropic Claude API
	Hour 3 – Lab	In-Class Lab: Build a Streamlit market intelligence dashboard (VS Code) with — stock screener, candlestick charts (Plotly), portfolio analytics panel, and FinBERT sentiment feed from NSE news; integrate real-time data (yfinance); peer review for	Design and deploy an interactive financial dashboard with live data and NLP feeds (L6, L5)	In-Class Python Lab (VS Code + Streamlit + Plotly + yfinance + FinBERT); Live Demo; Peer Review	Streamlit Documentation; Plotly Python Docs; yfinance Library

Sess.	Hour	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		usability and data accuracy			
Session 10 Module 10: GCC Capstone – Integrated Financial AI System 3 Hours					
10	Hour 1	Capstone Integration: Combine all module deliverables — RAG prospectus bot, Monte Carlo VaR engine, Basel III disclosure generator, NAV automation, Streamlit dashboard — into a unified Integrated Financial AI System; system architecture documentation; integration testing; GitHub portfolio setup; capstone evaluation rubric walkthrough	Design integrated AI system; evaluate project readiness against GCC hiring criteria (L6, L5)	Workshop (Instructor-led integration session + Peer Review rounds); Integration Lab; GitHub Setup	McKinsey GCC India Report (2024); Jadhav, <i>Claude AI for Finance Professionals</i> (Capstone Chapter)
	Hour 2	Pitch Deck Design & Mock Rehearsal: Pitch deck structure for GCC hiring panels — Problem → Solution Architecture → Live Demo → Business Impact → Team; AI-assisted resume and cover letter tailoring; common technical interview questions (Python, finance, AI governance);	Design and deliver a GCC-standard project pitch; evaluate communication effectiveness (L6, L5)	Group Rehearsal; Structured Peer Feedback (GCC hiring rubric); Instructor Coaching Round	NASSCOM GCC Trends Report (2024); Course Portfolio Rubric

Sess.	Hour	Topic / Sub-Topic	Learning Objectives	Pedagogy & Activity	Teaching Aids & References
		structured peer rehearsal with GCC hiring rubric feedback			
	Hour 3 – Capstone	GCC Capstone Presentation: Live 15-minute project presentation to mock GCC hiring panel (EisnerAmper / Opus Fund Services scenario); 10-minute Q&A with panel; placement readiness scoring on technical depth, communication, portfolio quality, and role alignment; individual feedback; course debrief — industry outlook and next steps	Present integrated system to mock panel; demonstrate placement-ready communication (L6)	Mock Hiring Panel (faculty as GCC panel); Peer Evaluation Rubric; Course Debrief Discussion	NASSCOM GCC Report (2024); EisnerAmper GCC Case; Course Portfolio Rubric

Assessments

Assignment: AI-Powered Monte Carlo VaR Risk Report

Introduced in: Session 5 (Hour 2) | **Submitted to Academic Coordinator:** At least 2 sessions before due date

Conducted: In-class lab component (Session 5, Hour 3) + individual take-home writeup

Description: Students will develop a Python-based Monte Carlo Value-at-Risk (VaR) model for a hypothetical equity portfolio of five NSE-listed stocks. The assignment requires students to: (1) collect 2 years of historical price data using Yahoo Finance API in Python (VS Code), (2) run 10,000 Monte Carlo simulation paths using NumPy, (3) calculate 95% and 99% VaR and CVaR, (4) generate an AI-written risk commentary using Claude API, and (5) interpret results with reference to Basel III market risk capital requirements.

Deliverables:

- Python script (.py) with all code, charts, and simulation output (VS Code project)
- 2-page PDF summary report with AI-generated commentary and regulatory interpretation

Criterion	Model Accuracy & Code	AI Integration Quality	Regulatory Interpretation	Report Presentation	Total
Marks	30	30	20	20	100

Case Study: "EisnerAmper GCC Pune – Automating Audit Analytics with Generative AI"

Conducted: Session 7, Hour 2 (in-class, 45 minutes discussion + 30 minutes written response)

Submitted to Academic Coordinator: Case material distributed and submitted to coordinator at least 2 sessions before the session

Case Summary: EisnerAmper's Global Capability Centre in Pune adopted generative AI tools to automate document review, extract financial data from unstructured PDFs, and draft preliminary audit findings. The case explores: technology stack selection, change management challenges, regulatory risk (PCAOB/ICAI audit standards), ROI measurement, and competitive implications for mid-tier audit firms entering the GCC model.

Discussion Questions:

- What specific audit tasks were automated, and which AI techniques were applied to each?
- Analyse the governance and data privacy risks EisnerAmper faced in deploying AI in audit workflows.
- Using the AI governance framework from Session 7, evaluate EisnerAmper's approach. What controls were missing?
- How does this GCC model create competitive moats, and what does it mean for entry-level finance careers?

Written Submission: Individual 1-page analytical response (submitted within 48 hours of class)

Criterion	Case Comprehension	Analytical Depth	Framework Application	Written Communication	Total
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Marks	25	35	25	15	100
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References

Textbooks (Primary)

1. Hilpisch, Y. (2020). *Artificial Intelligence in Finance: A Python-Based Guide*. O'Reilly Media, Sebastopol, CA. ISBN: 978-1492055433. **Textbook**
2. Chan, E. P. (2013). *Algorithmic Trading: Winning Strategies and Their Rationale*. Wiley Trading, Hoboken, NJ. ISBN: 978-1118460146. **Textbook**
3. Hull, J. C. (2022). *Options, Futures, and Other Derivatives* (11th ed.). Pearson Education, Upper Saddle River, NJ. ISBN: 978-0136939917. **Textbook**
4. Jadhav, T. (2024). *Claude AI for Finance Professionals: Agentic Workflows, RAG, and Automation*. Amazon KDP. [Amazon Best Seller – AI & Finance Category] **Textbook**

Research Papers (Primary)

1. Yang, Z., Dai, Z., Yang, Y., Carbonell, J., Salakhutdinov, R., & Le, Q. V. (2019). *XLNet: Generalized Autoregressive Pretraining for Language Understanding*. NeurIPS 2019. ArXiv: 1906.08237. **Research Paper**
2. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., ... Kiela, D. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. NeurIPS 2020. ArXiv:2005.11401. **Research Paper**
3. White, J., Fu, Q., Hays, S., Sandborn, M., Olea, C., Gilbert, H., ... Schmidt, D. C. (2023). *A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT*. ArXiv:2302.11382. **Research Paper**
4. Financial Stability Board (FSB). (2022). *Artificial Intelligence and Machine Learning in Financial Services: Market Developments and Financial Stability Implications*. FSB Report, November 2022. **Research Paper**

Institutional & Regulatory References

1. Basel Committee on Banking Supervision. (2017). *Basel III: Finalising Post-Crisis Reforms*. Bank for International Settlements (BIS), December 2017.
2. GARP. (2024). *FRM Exam Part II Study Guide: Market Risk Measurement and Management*. Global Association of Risk Professionals.
3. NASSCOM. (2024). *GCC in India: Talent, Technology and Transformation*. NASSCOM Research Report.
4. Federal Reserve System. (2011). *SR 11-7: Guidance on Model Risk Management*. Board of Governors of the Federal Reserve System.

90 Relevant Topics – Specialization Subject List

As required under Guideline 7: Submitted by 13th August 2026. Topics span all 10 modules and adjacent industry domains.

1. Transformer architecture and attention mechanism	2. Pre-training and fine-tuning of LLMs	3. Tokenization and embedding models	Category A: Large Language Models & AI Foundation (Topics 1–9)
6. Open-source vs. proprietary LLMs in finance	7. Multimodal AI in financial document analysis	8. AI-assisted financial research and report generation	
		4. LLM hallucination and mitigation strategies	
			9. FinBERT domain-specific language models
10. Zero-shot prompting in financial contexts	11. Few-shot and chain-of-thought prompting	12. Structured output prompting (JSON / tables)	Category B: Prompt Engineering & RAG (Topics 10–18)
15. Document chunking and embedding strategies	16. RAG pipeline evaluation metrics	17. DRHP and prospectus analysis using RAG	
		13. System prompt design for financial agents	
			18. Agent-based RAG vs. standard RAG architecture

19. Agent loop: observe–plan–act–reflect	20. Model Context Protocol (MCP) specification	21. Tool calling and function execution in agents	Category C: Agentic AI & Orchestration (Topics 19–27)
24. Investment memo generation pipeline (3-agent)	25. Autonomous financial research agents	26. Human-in-the-loop agent design	
			22. Multi-agent framework: CrewAI and LangGraph
			27. Agent evaluation: quality assurance
28. Market microstructure and order book dynamics	29. Momentum and mean-reversion trading strategies	30. Pairs trading and statistical arbitrage	Category D: Algorithmic Trading (Topics 28–36)
33. Reinforcement learning for trading (DQN, PPO)	34. NLP-based trading signal generation	35. High-frequency trading (HFT) and latency	
			31. Backtesting methodologies and overfitting risks
			36. SEBI algorithmic trading regulations
37. Value-at-Risk (VaR): historical and parametric	38. Monte Carlo simulation for market risk	39. Conditional Value-at-Risk (CVaR / Expected Shortfall)	Category E: Risk Analytics & Modelling

			(Topics 37–45)
42. Credit scorecard development with ML	43. Stress testing and scenario analysis	44. Model validation: Gini, KS statistic, AUC-ROC	40. Probability of Default model development 45. AI-generated commentaries and reports
46. Basel III capital adequacy framework (CET1, Tier 1)	47. Liquidity Coverage Ratio (LCR) and NSFR	48. Pillar 3 disclosure automation	Category F: RegTech & Compliance Automation (Topics 46–54)
51. FATF recommendations and RegTech compliance	52. Automated SAR (Suspicious Activity Report) filing	53. AxiomSL and Wolters Kluwer OneSumX platforms	49. Anti-Money Laundering (AML) and detection 54. RBI guidelines in banking
55. Model Risk Management (SR 11-7 framework)	56. Explainable AI (XAI): SHAP and LIME methods	57. Algorithmic bias and fairness in credit decisions	Category G: AI Governance & Ethics (Topics 55–63)

60. Data privacy in AI: GDPR and DPDP Act (India)	61. AI Total Cost of Ownership (TCO) modelling	62. Responsible AI principles for financial institutions	58. EU AI Act and financial services implications
			63. AI audit model documentation standards
64. Net Asset Value (NAV) calculation lifecycle	65. Corporate actions processing and automation	66. Transfer Agency operations and e-KYC	Category H: Fund Administration & FinOps (Topics 64–72)
69. Reconciliation automation in fund operations	70. Alternative investment fund (AIF) administration	71. SEBI mutual fund regulations and compliance	
			67. Fund accounting standards 9, ASC 820
			72. AI in portfolio performance attribution
73. Sentiment analysis with FinBERT	74. Earnings call transcript mining	75. News-based event-driven trading signals	Category I: Market Analytics & Visualization (Topics 73–81)
78. ESG data integration in valuation models	79. Streamlit dashboard development for finance	80. Plotly and interactive financial charting	
			76. Discounted Cash Flow (DCF) model with AI

				81. Real-time market data APIs (NSE, BSE, FRED)
82. Global Capability Centres (GCC): India landscape 2024	83. GCC hiring criteria in finance and AI roles	84. Portfolio design for finance + AI job applications	Category J: GCC Operations Capstone & Career Readiness (Topics 82–90)	
87. Technical interview preparation: Python + Finance	88. Capstone: Integrated Financial AI System design	89. Industry mentorship and alumni network leverage		
				85. Pitch design for technical finance product
				90. Emerging trends: Quantum computing in finance

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Coordinator

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